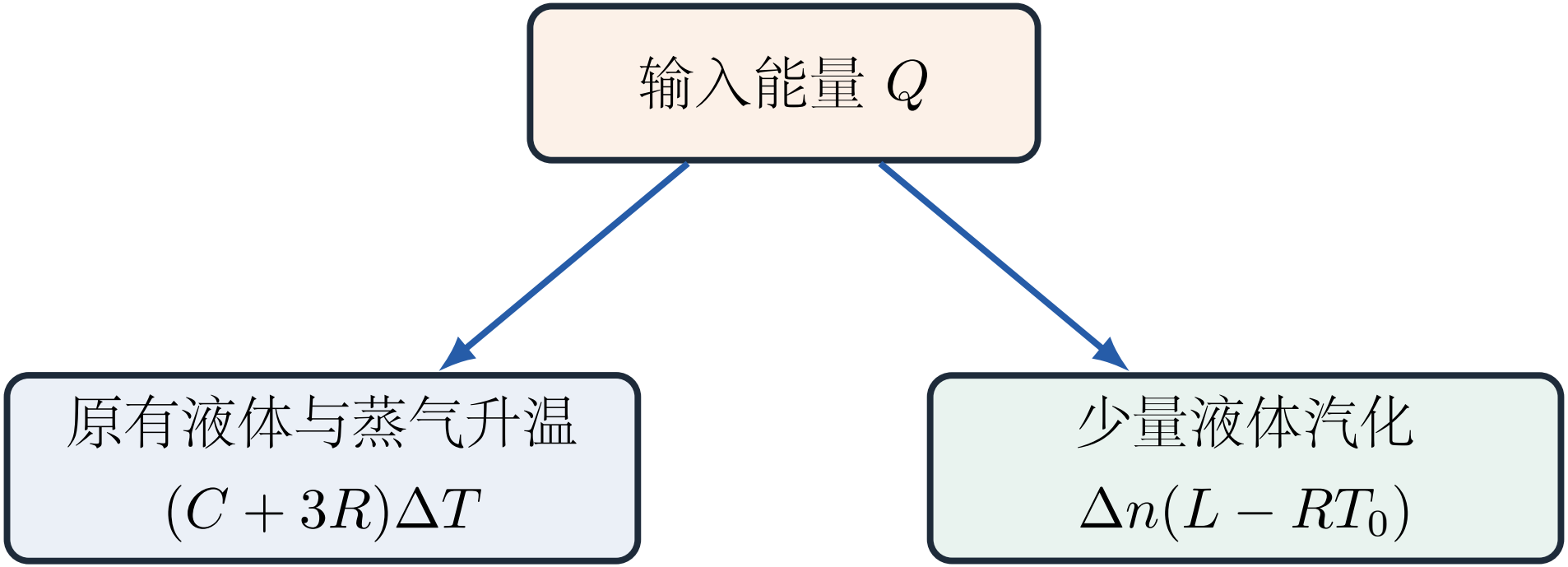

$$\Delta n = \frac{n_0 \Delta T}{T_0}\left(\frac{L}{RT_0} - 1\right)$$

$$Q = (C+3R)\Delta T + \Delta n(L - RT_0)$$